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RESEARCH

Two trajectories of lumbar disc degeneration from adolescence to adulthood: a 23-year longitudinal MRI study

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Abstract

Purpose Disc degeneration (DD) increases with age, but its onset and early progression remain poorly understood. This longitudinal study examined DD progression from adolescence into adulthood in healthy individuals.

Methods Forty participants with magnetic resonance imaging (MRI) at ages 11, 19, and 34 were included. Discs were graded using the Pfirrmann classification, and a Pfirrmann Summary Score (PSS, range 5–25) was calculated by summing individual disc grades. The primary outcome was progression of overall DD burden, defined as an increase in PSS. Participants were grouped by PSS at age 11: PSS = 5 (later-onset) vs. PSS 6 or higher (early-onset). Presence of at least one disc graded Pfirrmann 3 or higher was assessed at ages 19 and 34.

Results Demographics, anthropometrics, and clinical characteristics were comparable between the groups. All participants showed DD progression. Early-onset individuals had higher PSS at ages 19 and 34, whereas later-onset individuals exhibited greater PSS change over time. By age 19, early-onset participants more frequently had at least one Pfirrmann 3 or higher disc; by age 34, occurrence was similar (72% vs. 73%). Later-onset DD was localized to lower lumbar spine, whereas early-onset DD was more widespread. DD progression trajectories were not associated with low back pain (LBP) by age 34.

Conclusion Progression of DD is an inevitable but individually variable process. Early-onset DD appears to lead to a more widespread pattern, whereas later-onset DD progresses more rapidly but remains confined to the lower lumbar spine. No direct relationship between structural progression and LBP could be determined.

Keywords Disc degeneration · Progression · Magnetic resonance imaging · Longitudinal · Adolescence · Adulthood

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Introduction

In young, otherwise healthy populations, disc degeneration (DD) is more prevalent than most other degenerative conditions [1]. Magnetic resonance imaging (MRI) studies have demonstrated early signs of DD in approximately 11% of children under the age of 7 [2], with the occurrence increasing to 25% by the age of 10 [3]. The pubertal growth spurt appears to represent a critical period for DD development, as nearly half of individuals in their early 20s already exhibit DD [4]. A potential genetic contribution to early-onset DD has been suggested [5].

The lifetime trajectory of DD remains poorly characterized and likely reflects multiple contributing factors. Several phenotypes have been described [6], possibly representing distinct pathways of degeneration [7]. Age is a consistent risk factor for DD progression: both the proportion of affected individuals and the number of involved levels increase with advancing age [6]. However, progression

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during adulthood is generally slow, with population-based data suggesting annual progression rates of 13–15% [8]. Baseline DD strongly predicts subsequent progression [9], which appears most pronounced in younger adults and tends to slow after age 50 [10]. Environmental influences such as mechanical loading and smoking contribute to progression, but twin studies indicate a substantial genetic component [11, 12], with heritability estimates reaching up to 76% before age 50 [12].

Only a few longitudinal studies have examined the early progression of DD. In adolescents aged 12–14 years, progression occurred in 16% of participants over 2.7 years [13]. In young men with low back pain (LBP) at baseline, the proportion of degenerated discs increased from 21% at age 20 to 60% at age 37 [14]. Among women in their early 20s, 52% showed progression over a 10-year follow-up, predominantly in the lower lumbar spine [15]. Across these studies, baseline DD consistently predicted future progression.

Taken together, current evidence suggests that early-onset DD may be an important determinant of later spinal health, yet its long-term natural history remains unclear. Therefore, the present study aimed to characterize the natural course of lumbar DD from early adolescence to adulthood in a 23-year longitudinal cohort of healthy volunteers. Specifically, we sought to (1) compare the long-term progression of DD between participants with and without early-onset DD, (2) determine the anatomical distribution of DD progression across lumbar levels, and (3) examine the relationship between DD progression and the lifetime occurrence of low back pain (LBP) in adulthood.

Materials and methods

Study design and participants

A longitudinal cohort was established in 1994 to investigate lumbar spine development in healthy children. A random sample of 8-year-old students with even-numbered birth dates was recruited from six primary schools in the Helsinki metropolitan area ($n=94$ at baseline from an eligible population of $n=208$). Follow-up assessments were conducted at ages 11 ($n=81$) and 19 ($n=71$). In 2021, all traceable members of the original cohort ($n=89$) were recontacted by mail, yielding a final follow-up sample of 48 individuals (51% of the baseline cohort) at age 34. For the present study, we included 40 participants with complete data at ages 11, 19, and 34, including a structured interview, clinical examination, and lumbar spine MRI.

Structured interview and clinical examination

Structured interviews covered physical activity and sports participation, smoking (where age-appropriate), history of LBP unrelated to specific trauma (during the last week, last month, last year, earlier), and any healthcare use for LBP. Clinical measures included anthropometry for the calculation of Body Mass Index (BMI; ISO-BMI at ages 8 and 11, standard adult formula thereafter), and Body Surface Area as a two-dimensional measure of body size (Mosteller formula: square root of height in centimeters multiplied by weight in kilograms divided by 3600). A standard musculoskeletal clinical examination was also performed.

At the final follow-up, additional validated patient-reported outcome measures (PROMs) were collected: Numeric Rating Scale (NRS, 0–10) for back, neck, arm, and leg pain over the past week; Oswestry Disability Index (ODI) for spine-related disability; EQ-5D-5L for health-related quality of life; International Physical Activity Questionnaire - Short Form (IPAQ-SF) for physical activity.

MRI acquisition and evaluation

Lumbar spine MRI was performed using 1.0 T high-field scanners at ages 8 and 11 and 1.5 T high-field scanners at ages 19 and 34, all equipped with dedicated spine coils. Early imaging (ages 8, 11, 19) included sagittal T2-weighted sequences, while the final imaging protocol also incorporated sagittal T1-weighted, STIR, and axial T2-weighted sequences. To minimize potential diurnal variation in disc signal intensity, all scans were scheduled before 10:00 AM.

DD at levels L1-S1 was graded from sagittal midline T2-weighted images using the 5-point Pfirrmann classification [16]. Two independent evaluators, a musculoskeletal radiologist and a spine surgeon, assessed all images blinded to participants' clinical information and prior imaging. Inter-rater reliability (Cohen's kappa) ranged from 0.73 to 0.84, indicating substantial agreement [17]. Disagreements were resolved by consensus with a third evaluator.

To quantify the overall DD burden, a Pfirrmann Summary Score (PSS) was calculated by summing the Pfirrmann grades (1–5) across all lumbar levels (range 5–25) [18]. Participants were grouped according to PSS at age 11: later-onset group (PSS = 5, all discs Pfirrmann grade 1) and early-onset group (PSS 6 or higher, early disc changes at one or more levels). At ages 19 and 34, discs with Pfirrmann grade 3 or higher were regarded as degenerated.

Outcomes

The primary outcome was progression of overall DD burden, defined as an increase in PSS compared with the previous assessment. We also analyzed the proportion of participants with at least one degenerated disc (Pfirrmann 3 or higher) in the entire lumbar spine and at individual levels at ages 19 and 34, and the relationship between DD progression and LBP.

Statistical analysis

Continuous variables are presented using mean and standard deviation (SD) and categorical variables using frequency and percentage of total. Statistical evaluation between groups (later-onset and early-onset) was analyzed by using Student's t-test, permutation test (Monte Carlo p-values) and Pearson's chi-squared test. Repeated measures were analyzed using permutation type generalizing estimating equations (GEE) models. A permutation type test was used because the number of observations in each group was small, and the assumptions underlying the corresponding parametric test could not be relied upon. Confidence intervals for the means were obtained by bias-corrected bootstrapping (5000 replications). The Stata 18, StataCorp LP (College Station, TX, USA) statistical package was used for the above analyses.

This study followed the STROBE guidelines for reporting observational research.

Results

For the present study, we analyzed data from 40 participants with complete data at ages 11, 19, and 34.

Baseline characteristics

Table 1 summarizes the demographic, anthropometric, and clinical characteristics of the two groups at ages 11 and 34. No statistically significant between-group differences were observed at either baseline or final follow-up. By age 11, participants with later-onset and early-onset disc changes had achieved, on average, 87% and 89% of their adult height, and 54% and 53% of their adult weight, respectively.

Progression of disc changes

Figure 1 shows the distribution of PSS values at ages 11, 19, and 34 according to baseline PSS. Progression of DD was observed in all participants over the 23-year follow-up. In the early-onset group, mean (SD) PSS values were 7.2 (1.2), 10.9 (1.4), and 12.2 (2.0) at ages 11, 19, and 34, respectively. In the later-onset group, the corresponding values were 9.7 (1.4) at age 19 and 10.7 (0.8) at age 34. Between-group differences were statistically significant at ages 19 and 34.

Figure 2 illustrates the mean change in PSS over time in both groups. Significant progression occurred at all study time points. Pronounced between-group differences emerged at ages 19 and 34, with participants in the later-onset group showing a greater mean increase in PSS over time compared to the early-onset group.

Occurrence of disc degeneration

Table 2 presents the occurrence of at least one Pfirrmann 3 or higher disc across the lumbar spine and at individual levels. At age 19, DD was significantly more frequent in

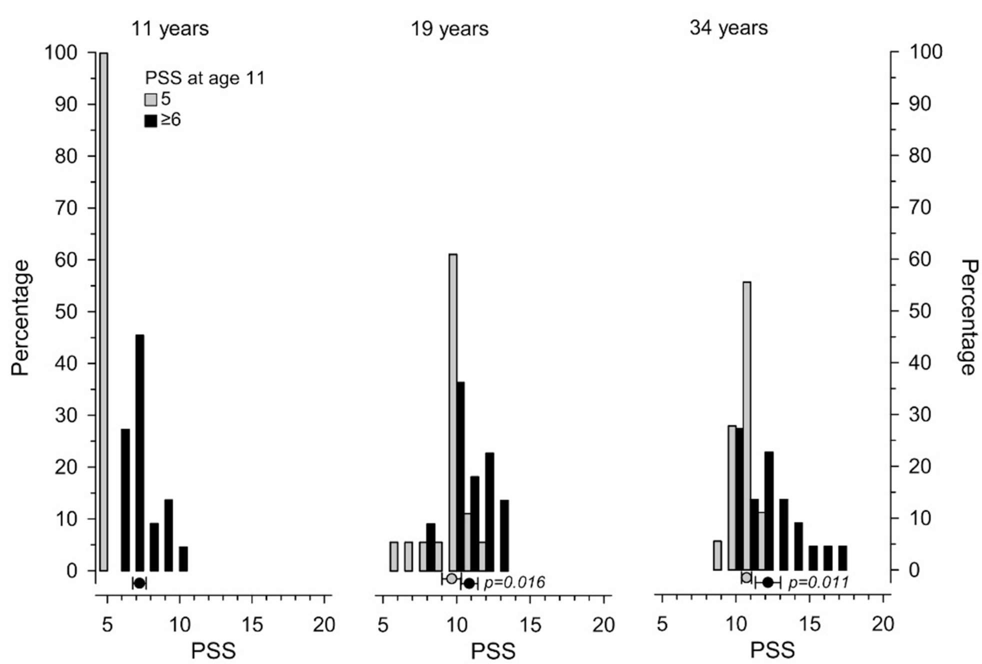

Fig. 1 Distribution of Pfirrmann Summary Score (PSS) values at ages 11, 19 and 34 according to baseline PSS. Circles represent mean values and error bars are for 95% confidence intervals

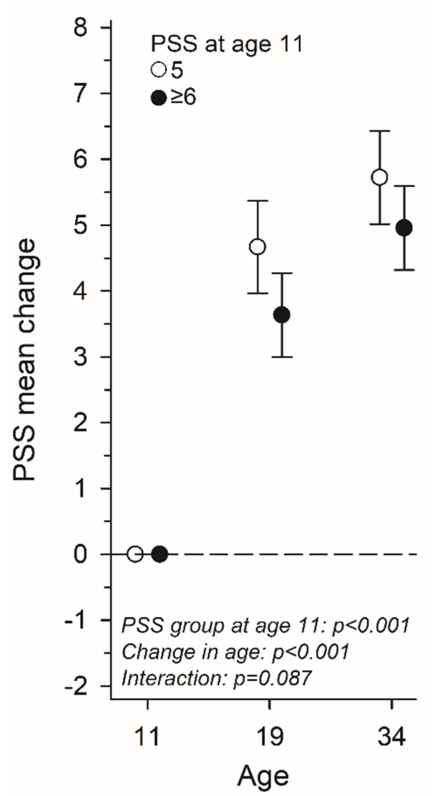

Fig. 2 Mean change in Pfirrmann Summary Score (PSS) over time according to baseline PSS. Error bars are for 95% confidence intervals

Table 2 Participants with at least one Pfirrmann 3 or higher disc for the entire lumbar spine and the individual lumbar levels according to baseline Pfirrmann summary score (PSS)

Level/s	PSS = 5 n = 18	PSS 6 or higher n = 22	p-value
	n (%)	n (%)	
Age 19			
L 1–5	5 (28)	15 (68)	0.025
L1	0 (0)	2 (9)	
L2	0 (0)	2 (9)	
L3	0 (0)	1 (5)	
L4	2 (11)	5 (23)	
L5	3 (17)	12 (55)	
Age 34			
L 1–5	13 (72)	16 (73)	0.99
L1	0 (0)	2 (9)	
L2	0 (0)	2 (9)	
L3	0 (0)	4 (18)	
L4	6 (33)	11 (50)	
L5	9 (50)	12 (55)	

the early-onset group. By age 34, the occurrence was nearly identical between groups. In the later-onset group, DD was localized to the lower lumbar spine, whereas in the early-onset group, it was more widely distributed. In the latter group, DD at L5-S1 remained stable from age 19 to 34, while progression was observed at L4-L5 and L3-L4. The upper lumbar levels (L1-L2 and L2-L3) showed minimal change over time.

Lower lumbar spine (L4-L5 and L5-S1)

At age 19, 28% of participants in the later-onset group and 59% of those in the early-onset group had at least one lower lumbar disc graded Pfirrmann 3 or higher. By age 34, these proportions had increased to 72% and 68%, respectively.

Low back pain

There was no difference between the two groups in the lifetime occurrence of LBP at age 34 (Table 1). Moreover, no significant relationship was found between LBP by age 34 and the mean change in PSS over time (Fig. 3).

Discussion

In this 23-year longitudinal study, all participants demonstrated progression of DD from early adolescence into adulthood, although the extent and trajectory varied between individuals. Disc changes appeared to accelerate during adolescence and decelerate in early adulthood. Two distinct trajectories were observed: participants with baseline disc changes maintained higher PSS throughout follow-up, whereas those without baseline changes showed greater progression over time.

When DD was defined as the presence of at least one disc graded Pfirrmann 3 or higher, the early-onset group showed a significantly higher occurrence at age 19. By age 34, however, this difference had converged, with nearly three-quarters of participants affected regardless of baseline status. Across both groups, the lower lumbar spine (L4-L5 and L5-S1) consistently carried the greatest DD burden. Level-specific analyses indicated that by age 34, participants with early-onset changes tended to develop more widespread DD extending to the mid-lumbar levels, whereas those without baseline changes showed a more localized pattern confined to the lower lumbar spine.

Our current understanding of DD progression during adolescence and early adulthood comes from only a few longitudinal MRI studies. Prior research has reported annual progression in approximately 9% of adolescents over a 2.7-year follow-up [13]. In young men with LBP, the mean number of degenerated discs increased from 1.1 to 3.0 over 17 years [14], while in young women, progression over 10 years occurred in 58% and 31% of those with and without baseline DD, respectively [15]. The present study extends these findings by spanning the period from early

Table 1 Characteristics of the study groups according to baseline Pfirrmann summary score (PSS)

Variable	PSS at age 11 5 $n=18$	6 or higher $n=22$	*p*-value
Women, n (%)	11(61)	9 (41)	0.20
Age 11			
ISO-BMI, mean (SD)	20.9 (3.4)	21.7 (3.8)	0.47
Body Surface Area, mean (SD)	1.30 (0.10)	1.37 (0.20)	0.17
Sitting height, mean (SD)	77.3 (2.8)	79.8 (5.3)	0.067
Height, mean (SD)	150 (5)	154 (10)	0.23
Weight, mean (SD)	40.5 (6.1)	44.6 (11.2)	0.18
LBP, n (%)	2 (11)	3 (14)	0.99
Physical activity, n (%)	14 (78)	14 (64)	0.49
Finger-floor-distance, mean (SD)	4.9 (6.5)	5.7 (7.5)	0.72
Age 34			
BMI, mean (SD)	25.3 (4.1)	28.0 (6.3)	0.13
Body Surface Area, mean (SD)	1.89 (0.17)	2.00 (0.24)	0.12
Height, mean (SD)	172 (10)	173 (10)	0.82
Weight, mean (SD)	75 (11)	84 (19)	0.086
LBP, n (%)	11 (61)	18 (82)	0.17
Physical activity, n (%)	13 (72)	19 (86)	0.43
Finger-floor-distance, mean (SD)	1.5 (3.3)	3.5 (5.9)	0.20
Smoking, n (%)			0.30
Never	10 (56)	13 (59)	
Quitted	3 (17)	7 (32)	
Smoking	5 (28)	2 (9)	
ODI, mean (SD)	23.9 (2.5)	25.7 (3.4)	0.078
EQ-5D-5L, mean (SD)	0.930 (0.103)	0.863 (0.133)	0.10
NRS LBP, mean (SD)	1.2 (1.6)	1.2 (1.3)	0.92
METhour/week, mean (SD)	78 (90)	61 (59)	0.49
Employed, n (%)	16 (89)	22 (100)	0.20
Non-manual occupation[a], n (%)	10 (59)	14 (63)	0.76
Physical workload[b], mean (SD)	10.5 (5.6)	8.5 (5.5)	0.27

[a] Participants were categorized into manual and non-manual (white-collar) occupations based on job descriptions

[b] Physical workload was evaluated using a 5-point scale across four categories: lifting and/or carrying heavy loads, repetitive physical tasks, prolonged sitting and/or standing, awkward and/or static postures. A total score ranging from 0 to 20 was calculated, with higher values indicating greater physical workload

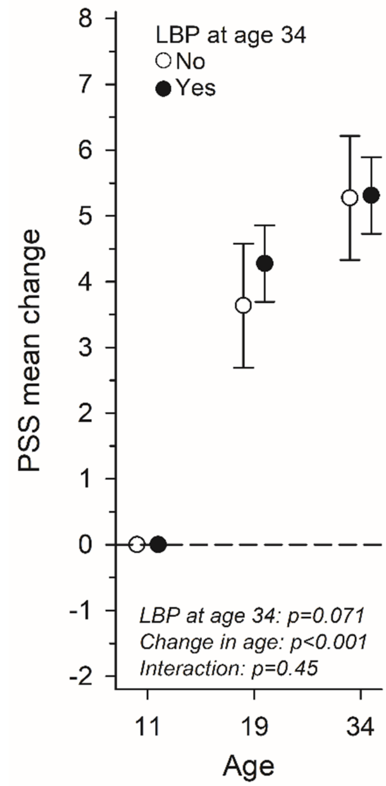

Fig. 3 Mean change of Pfirrmann Summary Score (PSS) over time according to LBP status by age 34

adolescence through the fourth decade of life and by suggesting two distinct trajectories of DD progression.

Taken together, our findings indicate that the onset of DD is highly variable: some individuals develop detectable changes before puberty, while others remain unaffected until the pubertal growth spurt or later. Baseline disc changes appeared to predict a more widespread degenerative trajectory, yet participants with initially normal discs demonstrated greater progression over time. While the rate and regional pattern of disc changes differed between individuals, progression itself appeared largely inevitable. Our data cannot directly address underlying mechanisms, but the observed patterns are compatible with the notion that "age-related" and "pathological" DD may not represent distinct entities, but rather a continuum within the same biological process [19]. Individual variability may therefore reflect differences in the timing of onset and rate of progression rather than fundamentally different mechanisms.

Some of the observed progression might theoretically reflect regression to the mean [20]. Participants were grouped based on their baseline PSS, with PSS = 5 indicating that all discs were graded Pfirrmann 1. Although this threshold is somewhat arbitrary, it cannot be considered extreme, as Pfirrmann grade 1 is generally regarded as normal in children and adolescents [21]. Consistent with our findings, prior longitudinal studies indicate that less degenerated discs have a greater potential for change, whereas more advanced degeneration progresses slowly [12, 22, 23]. This pattern may partly explain why participants with initially normal discs showed greater progression over time.

LBP became more common by age 34 in both groups. Despite differences in the onset and progression of DD, the lifetime occurrence of self-reported LBP by age 34 did not differ significantly between groups. Prior research suggests that rapid DD progression, rather than static cross-sectional findings, may be more relevant to symptom development [12]. In this cohort, the mean change in PSS over time was similar in participants with and without LBP by age 34, underscoring that the relationship between structural degeneration and clinical symptoms is not straightforward.

Several limitations should be acknowledged. The relatively small sample size (40 of 94 original participants) reduced statistical power and limited subgroup analyses, such as the identification of additional phenotypes [6]. MRI was performed using different scanners over time, but all were high-field systems with dedicated spine coils and optimized protocols. The present analysis focused specifically on the progression of DD, without assessing other morphological spinal changes. Composite measures such as the PSS do not differentiate between localized severe changes and widespread mild changes, but their use has been recommended, especially when associating imaging findings with clinical symptoms [18]. Moreover, the Pfirrmann classification itself is subjective, compresses a multidimensional degenerative process into a five-point ordinal scale, and does not capture early biochemical alterations that may precede structural changes [24]. Consequently, some participants in the later-onset group may have had early biochemical disc changes not detectable on MRI at baseline.

This study also has several notable strengths. It followed a well-defined cohort of healthy children with repeated MRI assessments spanning more than two decades. Recruitment was independent of symptoms, enhancing generalizability to otherwise healthy populations. The longitudinal design enabled characterization of individual DD trajectories and their evolution over time, providing novel insights into the natural history of DD that would not be captured in cross-sectional studies.

Conclusions

Taken together, our findings confirm that the progression of DD is an inevitable, yet individually variable process. Two distinct developmental trajectories appear to emerge: one characterized by early-onset, more widespread disc changes, and another by later-onset, more localized changes accompanied by a greater increase in overall degeneration burden (mean PSS change) over time.

The relationship between structural degeneration and clinical symptoms remains complex and not straightforward. Larger longitudinal cohorts with detailed symptom data are needed to validate these developmental patterns and clarify their clinical significance, particularly in relation to pain and functional outcomes. Importantly, early identification of disc changes may offer opportunities for preventive or therapeutic strategies, even if the symptomatic impact during youth remains uncertain to date.

Author contributions Conseptualisation: T.L. and D.S. Data curation: L.R. Formal analysis: M.L., D.S., H.K., L.R., A.A., T.L. Investigation: A.A. Methodology: M.L., D.S., H.K., T.L. Project administration: L.R. Writing-original draft: T.L. Figures 1, 2 and 3: H.K. All authors read and approved the final manuscript.

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Data availability The datasets generated and/or analyzed during the current study are available from the corresponding author upon reasonable request and in accordance with an approved protocol, provided that current legislation permits it.

Declarations

Competing interests The authors declare no competing interests.

Ethics approval The study followed the Declaration of Helsinki. Ethical approval for the original study was granted by the Ethics Committee of The Invalid Foundation in 1993, with additional permission from the Helsinki school district. The final follow-up protocol was approved by the Ethics Committee of Helsinki University Hospital (protocol 3251/2020). Written informed consent was obtained from legal guardians at baseline and from all individual participants at final follow-up.

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